

$$(A+B)^T = \begin{bmatrix} (a_{11}+b_{11}) & (a_{12}+b_{12}) & (a_{13}+b_{13}) & \dots & (a_{1n}+b_{1n}) \\ (a_{21}+b_{21}) & (a_{22}+b_{22}) & (a_{23}+b_{23}) & \dots & (a_{2n}+b_{2n}) \\ (a_{31}+b_{31}) & (a_{32}+b_{32}) & (a_{33}+b_{33}) & \dots & (a_{3n}+b_{3n}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (a_{m1}+b_{m1}) & (a_{m2}+b_{m2}) & (a_{m3}+b_{m3}) & \dots & (a_{mn}+b_{mn}) \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{21} & a_{31} & \dots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \dots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \dots & a_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \dots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{21} & b_{31} & \dots & b_{m1} \\ b_{12} & b_{22} & b_{32} & \dots & b_{m2} \\ b_{13} & b_{23} & b_{33} & \dots & b_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{1n} & b_{2n} & b_{3n} & \dots & b_{mn} \end{bmatrix}$$

(By defn. of transpose)

$$= A^T + B^T \quad (\text{By defn. of matrix addition})$$

$$\therefore (A+B)^T = A^T + B^T \quad \square$$

$$c) \quad rA = r \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} ra_{11} & ra_{12} & ra_{13} & \dots & ra_{1n} \\ ra_{21} & ra_{22} & ra_{23} & \dots & ra_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ ra_{m1} & ra_{m2} & ra_{m3} & \dots & ra_{mn} \end{bmatrix}$$

(By defn. of scalar multiple)

$$(rA)^T = \begin{bmatrix} ra_{11} & ra_{21} & ra_{31} & \dots & ra_{m1} \\ ra_{12} & ra_{22} & ra_{32} & \dots & ra_{m2} \\ ra_{13} & ra_{23} & ra_{33} & \dots & ra_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ ra_{1n} & ra_{2n} & ra_{3n} & \dots & ra_{mn} \end{bmatrix} = r \begin{bmatrix} a_{11} & a_{21} & a_{31} & \dots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \dots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \dots & a_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \dots & a_{mn} \end{bmatrix}$$

(By defn. of transpose)

(By defn. of scalar multiple)

$$= rA^T \quad (\text{By defn. of transpose})$$

$$\therefore (rA)^T = rA^T \quad \square$$

For (d) we shall do it in Exercise 33.